

IDENTIFYING INFORMATION:

NAME: Bruce, Juliette

ORCID iD: <https://orcid.org/0000-0001-9766-7896>

POSITION TITLE: Assistant Professor

PRIMARY ORGANIZATION AND LOCATION: Dartmouth College, Hanover, NH, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Brown University, Providence, Rhode Island, United States	Postdoctoral Fellow	08/2022 - 06/2024	Mathematics
University of California, Berkeley, Berkeley, California, United States	Postdoctoral Fellow	08/2020 - 07/2022	Mathematics
Mathematical Sciences Research Institute, Berkeley, California, United States	Postdoctoral Fellow	08/2020 - 05/2021	Mathematics
University of Wisconsin–Madison, Madison, Wisconsin, US	Ph.D	09/2014 - 08/2020	Department of Mathematics
University of Wisconsin–Madison, Madison, Wisconsin, US	Masters of Arts	09/2014 - 05/2016	Department of Mathematics
University of Michigan–Ann Arbor, Ann Arbor, Michigan, US	Bachelors of Science	09/2010 - 04/2014	Department of Mathematics / Department of Political Science

Appointments and Positions

2024 - present Assistant Professor, Dartmouth College, Hanover, NH, United States

2025 - 2025 Long-Term Visiting Member, Fields Institute, Toronto, Ontario, Canada

2024 - present Assistant Professor, Dartmouth College, Mathematics, Hanover, New Hampshire, United States

2014 - 2020 Teaching Assistant / Research Assistant, University of Wisconsin - Madison, Department of Mathematics, Madison, WI, US

Products**Products Most Closely Related to the Proposed Project**

1. Madeline Brandt, Juliette Bruce, Melody Chan, Margarida Melo, Gwyneth Moreland, Corey Wolfe. On the top-weight rational cohomology of $\mathcal{A}_g$. Geometry & Topology.

2024 March. DOI: 10.48550/ARXIV.2012.02892

2. Bruce J, Erman D. A probabilistic approach to systems of parameters and Noether normalization. *Algebra & Number Theory*. 2019 December 7; 13(9):2081-2102. Available from: <https://msp.org/ant/2019/13-9/p05.xhtml> DOI: 10.2140/ant.2019.13.2081
3. Bruce J. Asymptotic syzygies for products of projective spaces. *Journal of Algebra*. 2024 July; 649:347-391. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0021869324001479> DOI: 10.1016/j.jalgebra.2024.03.013
4. Bruce J, Cranton Heller L, Sayrafi M. Characterizing multigraded regularity and virtual resolutions on products of projective spaces. *Advances in Mathematics*. 2026 August; 500:111049-. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0001870826002719> DOI: 10.1016/j.aim.2026.111049
5. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi. Bounds on Multigraded Regularity. 2022. Available from: <https://arxiv.org/abs/2208.11115> DOI: 10.48550/ARXIV.2208.11115

Other Significant Products, Whether or Not Related to the Proposed Project

1. Bruce J, Erman D, Goldstein S, Yang J. Conjectures and Computations about Veronese Syzygies. *Experimental Mathematics*. 2018 June 21; 29(4):398-413. Available from: <https://www.tandfonline.com/doi/full/10.1080/10586458.2018.1474506> DOI: 10.1080/10586458.2018.1474506
2. Juliette Bruce. The quantitative behavior of asymptotic syzygies for Hirzebruch surfaces. *Journal of Commutative Algebra*. 2022 March. DOI: 10.1216/jca.2022.14.19
3. Eran Assaf, Madeline Brandt, Juliette Bruce, Melody Chan, Raluca Vlad. Tropicalizations of locally symmetric varieties. 2025. Available from: <https://arxiv.org/abs/2505.10504> uri: <https://arxiv.org/abs/2505.10504>
4. Juliette Bruce. Eventual Nonstandard Koszulness Fails for Veronese Subrings of Weighted Polynomial Rings. 2026. Available from: <https://arxiv.org/abs/2608.23913> DOI: 10.48550/ARXIV.2608.23913
5. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi, Alexandra Secoleanu. Seshadri Regions and the Asymptotic Shape of Multigraded Regularity. 2025. Available from: <https://arxiv.org/abs/2512.05289> DOI: 10.48550/ARXIV.2512.05289

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Bruce, Juliette in SciENcv on 2026-09-10 04:05:46